

Common Distributions

The notation $z \sim P(\theta)$ means that the random variable z is sampled from (or distributed according to) the distribution P , which is parameterized by θ . When we write $P(z; \theta)$ we refer to the density (assuming it exists) of P evaluated at z . Here, we provide a summary of common distributions and their parametric densities or mass functions.

Bernoulli For a binary random variable $x \in \{0, 1\}$ with $\rho \in [0, 1]$,

$$\text{Bern}(x; \rho) = \rho^x (1 - \rho)^{1-x}.$$

Sometimes we parameterize the Bernoulli distribution in terms of its *logit*, $\sigma^{-1}(\rho) = \log \frac{\rho}{1-\rho}$, where $\sigma(u) = (1 + e^{-u})^{-1}$ is the *logistic/sigmoid* function.

Beta For a continuous random variable $\rho \in [0, 1]$ with $a > 0$ and $b > 0$,

$$\text{Beta}(\rho; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \rho^{a-1} (1-\rho)^{b-1}.$$

Note that for $\alpha < 1$ and/or $\beta < 1$, the beta density diverges and is **unbounded**.

Binomial For an integer-valued random variable $x \in \{1, \dots, N\}$ with $N \in \mathbb{N}$ and $\rho \in [0, 1]$,

$$\text{Bin}(x; N, \rho) = \binom{N}{x} \rho^x (1-\rho)^{N-x}.$$

Categorical For an integer-valued random variable $x \in \{1, \dots, K\}$ with $K \in \mathbb{N}$ and $\boldsymbol{\pi} \in \Delta_K$,

$$\text{Cat}(x; \boldsymbol{\pi}) = \pi_x,$$

where π_x denotes the x -th coordinate of the vector $\boldsymbol{\pi}$.

Note: sometimes the categorical distribution is used as a distribution on one-hot vectors $\mathbf{x} \in \{0, 1\}^K$; $\sum_{k=1}^K x_k = 1$, in which case the pmf is $\text{Cat}(\mathbf{x}; \boldsymbol{\pi}) = \prod_{k=1}^K \pi_k^{x_k}$. It will be clear from context if x is an integer or a one-hot vector.

Dirichlet For a probability vector $\boldsymbol{\pi} \in [0, 1]^K$ such that $\pi_k \geq 0$ and $\sum_k \pi_k = 1$, and parameter $\boldsymbol{\alpha} \in \mathbb{R}_+^K$,

$$\text{Dir}(\boldsymbol{\pi}; \boldsymbol{\alpha}) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k-1}.$$

Exponential For a random variable $x \in \mathbb{R}_+$ with rate $\lambda \in \mathbb{R}_+$,

$$\text{Exp}(x; \lambda) = \lambda e^{-\lambda x} \mathbb{I}[x \geq 0].$$

The cumulative distribution function for the exponential is $\Pr(x < a; \lambda) = 1 - e^{-\lambda a}$. You can sample an exponential distribution by first sampling $u \sim \text{Unif}([0, 1])$, then setting $x = -\frac{\log u}{\lambda}$.

Gamma For a nonnegative random variable $\lambda \in \mathbb{R}_+$ with shape parameter $a > 0$ and rate parameter $b > 0$,

$$\text{Ga}(\lambda; a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}.$$

The gamma distribution may also be parameterized in terms of a scale parameter, $\theta = b^{-1}$, but we do not use that parameterization in this exam.

Inverse Wishart For a random variable $\Sigma \in \mathbb{R}_{\geq 0}^{D \times D}$ (a positive semidefinite matrix) with degrees of freedom $\nu \in \mathbb{R}_+$ and scale $\Sigma_0 \in \mathbb{R}_{\geq 0}^{D \times D}$,

$$\text{IW}(\Sigma; \nu_0, \Sigma_0) = \frac{|\Sigma_0|^{\frac{\nu_0}{2}}}{2^{\frac{\nu_0 D}{2}} \Gamma_D\left(\frac{\nu_0}{2}\right)} |\Sigma|^{-\frac{\nu_0 + D + 1}{2}} e^{-\frac{1}{2} \text{Tr}(\Sigma_0 \Sigma^{-1})}.$$

Laplace (a.k.a. Double Exponential) For a random variable $x \in \mathbb{R}$ with rate $\lambda \in \mathbb{R}_+$,

$$\text{Lap}(x; \lambda) = \frac{\lambda}{2} e^{-\lambda |x|}.$$

Multinomial For a vector of discrete counts $\mathbf{x} \in \mathbb{N}^K$ with $\sum_k x_k = N$ and a probability vector $\boldsymbol{\pi} \in [0, 1]^K$,

$$\text{Mult}(\mathbf{x}; N, \boldsymbol{\pi}) = \binom{N}{x_1, x_2, \dots, x_K} \prod_{k=1}^K \pi_k^{x_k},$$

where

$$\binom{N}{x_1, x_2, \dots, x_K} = \frac{N!}{x_1! \dots x_K!}.$$

Multivariate Normal For a random variable $\mathbf{x} \in \mathbb{R}^D$ with mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and positive semidefinite covariance matrix $\Sigma \in \mathbb{R}^{D \times D}$,

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = (2\pi)^{-D/2} |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

Negative Binomial For an integer-valued random variable $x \in \mathbb{N}$ with shape parameters $\nu \in \mathbb{R}_+$ and probability $\rho \in [0, 1]$,

$$\text{NB}(x; \nu, \rho) = \binom{x + \nu - 1}{x} \rho^x (1 - \rho)^\nu.$$

Poisson For an integer random variable $x \in \mathbb{N}$ and a nonnegative rate parameters $\lambda \in \mathbb{R}_+$,

$$\text{Po}(x; \lambda) = \frac{1}{x!} \lambda^x e^{-\lambda}.$$

The mean and variance of the Poisson distribution are both equal to λ .

Truncated Normal For a random variable $x \in [a, b)$ and parameters $\mu \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}_+$,

$$\text{TruncNorm}(x; \mu, \sigma^2, [a, b)) = \frac{\mathcal{N}(x; \mu, \sigma^2)}{\int_a^b \mathcal{N}(y; \mu, \sigma^2) dy} \mathbb{I}[a \leq x < b].$$

Uniform For a continuous random variable $x \in [a, b)$,

$$\text{Unif}(x; [a, b)) = \frac{1}{b-a} \mathbb{I}[a \leq x < b].$$